Hamza Afzal

+92 3419696709; hamzacreed98@gmail.com;
GitHub https://github.com/hamzaafzal391 | LinkedIn https://www.linkedin.com/in/hamza-afzal-57b723340/

PROFESSIONAL SUMMARY

I'm an aspiring Data Scientist and AI engineer who loves working on real-world problems using data. I've worked on projects involving machine learning, time series forecasting, and both regression and classification tasks. I'm confident with Python, SQL, and data visualization, and always excited to learn more and build meaningful solutions.

EDUCATION

National University of Computer and Emerging Sciences (FAST-NUCES) ***2022 - 2026***
BS in Computer Science

- Relevant Courses: Linear Algebra(A), Differential Equations(A+), Probability and Statistics(A-), Calculus(A-), Database Systems (SQL) (A-)

Beaconhouse College Campus Gulberg ***2020 –2022***
A Levels

- Grades: Computer Science(**A***), Chemistry(**A***), Physics(**A***), Mathematics(**A***): High Achiever 100% Scholarship

PROJECTS

Air Quality Forecasting: Forecasted pollution trends using AR, ARMA, and SARIMA models on MongoDB time series data.
Earthquake Damage Analysis: Used SQL to extract features and applied classification models using Decision Trees and Logistic Regression to predict building damage severity levels.
Stroke Prediction: Used brain health data in Rapid Miner; ignored Random Forest despite highest accuracy, prioritizing recall-balanced Logistic Regression and Naive Bayes on imbalanced dataset.
Hospital Readmission Classification – SOFTEC ML Competition: Built a classification model on highly imbalanced hospital readmission data with missing values; topped Kaggle leaderboard in accuracy but placed runner-up overall due to recall-based evaluation.
Gas Price Prediction: Built SARIMA and Prophet models with Optuna tuning; SARIMA outperformed in forecasting gas prices accurately.
Railway Management System (SQL & ASP.NET): Designed a database-driven system with HTML/CSS/JS frontend for bookings, schedules, and passenger management.
Gig Generator Platform: Built a full-stack gig marketplace using Next.js and MongoDB, enabling users to post, manage, and browse tasks with dynamic routing and real-time updates.
Tetris (8088 Assembly): Created a Tetris clone with custom graphics, controls, and scoring logic using Assembly language.
Card Game (C++): Implemented a turn-based card game using OOP classes, managing game rules, interactions, and outcomes.

CERTIFICATIONS

Applied Data Science Lab – *WorldQuant University* (In Progress) 5/8 projects completed.
Data Analysis & Visualization – ***FreeCodeCamp***

EXPERIENCE

Data Science Summer Camp – LSE **July 2023 - August 2023**
Gained hands-on experience in machine learning and statistical modeling using Python, Pandas, NumPy, and Matplotlib, with a strong focus on real-world data applications.

FAST NUCES, Lahore – Teaching Assistant **January 2025 - June 2025**
Assisted 90+ students in DLD course, graded 100+ evaluations, and conducted personalized sessions to help students with DLD.

EXTRACURRICULARS

Campus Ambassador – Empath Ease: Represented a mental health organization on campus, promoting awareness and coordinating outreach initiatives.
Officer – SOFTEC: Managed logistics and operations for Pakistan's largest tech event across inventory, security, and EPC teams.
Member – AIESEC Pakistan: Led mental health workshops and recognized as Team Member of the Month for exceptional contribution.
Volunteer – WWF, CHCS, LUMS: Contributed to environmental and social causes through tree planting, food drives, and educational outreach.
Volunteer – Project Justuju (LUMS)
Taught underprivileged children in local schools, contributing to educational equity through weekly sessions.
Football Runner-Up – Misaal 2.0
Secured second place in an inter-university football tournament, showcasing teamwork and sportsmanship.

SKILLS

Languages: Python, C++, C, SQL, Assembly, HTML, CSS, Java, ASP.NET, C#
Tools, Libraries and Platforms: Pandas, NumPy, PyTorch, TensorFlow, Scikit-learn, Matplotlib, Seaborn, Plotly Express, RapidMiner, Tableau, MongoDB, Wireshark, MATLAB, Colab, Jupyter Notebook, Visual Studio, Visual Studio Code
Specializations: Data Analysis/Data Analytics, Data Visualization, Model Training, Time Series Forecasting, Regression/Classification Problems, OOP, Machine Learning
Soft Skills: Leadership, People Management, Excellent Communication

ACHIEVEMENTS

Runner-up, **SOFTEC** Machine Learning Competition (2025) – **Kaggle** leaderboard topper in hospital readmission prediction challenge